

A Note on One Authentication and Key Agreement Scheme for UAV-Assisted VANETs for Emergency Rescue

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Abstract. We show that the key agreement scheme [IEEE TNSE, 1454–1468, 2004] is insecure against ephemeral secret leakage attack and smart card loss attack, not as claimed. This failure results from its simple encryption, in which only bitwise XOR operations are used to encrypt the messages.

Keywords: Vehicular ad hoc networks, Key management, Unmanned aerial vehicles, Ephemeral secret leakage attack, Smart card loss attack.

1 Introduction

Recently, Cui et al. [2] have presented a key agreement protocol for Vehicular Ad hoc Networks (VANET). It is designed to achieve mutual authentication, session key agreement, anonymity, and resist password-guessing attack, impersonation attack, ephemeral secret leakage attack, smart card loss attack, etc. In this note, we show that the scheme cannot resist ephemeral secret leakage attack and smart card loss attack because of its simple encryption mechanism, in which only the bitwise XOR operations are involved.

2 Preliminary: bitwise XOR

The bitwise XOR, denoted by $\oplus$, compares two input bits and generates one output bit. When the operator is performed on two strings, the shorter string should be padded with some 0s to its left side. In this case, the partial string corresponding to the padding bits is eventually exposed. For example, the software, Mathematica, pads 0s to its left side by default. See the following code.

```
a = 161; b = 15;
IntegerString[{a, b, BitXor[a, b]}, 2]

{10100001, 1111, 10101110}
```

The partial string “1010” of the longer string “10100001” is directly copied into the final string “10101110”.

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3 Review of the Cui et al.'s key agreement

In the considered scenario, there are three entities. The trusted authority (TA) generates the system's public parameters and confidential keys. Each vehicle V_k is equipped with an on-board unit for communications. Each unmanned aerial vehicle (UAV) U_j , also known as a drone, is honest and curious. The scheme consists of seven phases: system setup, registration, login, authentication and key agreement, password update, re-registration, and honeywords storage optimization. The notations and descriptions are listed below (see Table 1).

Table 1: Notations and descriptions

V_k, U_j	k^{th} vehicle, j^{th} UAV
PW_k, RID_k	password, identity of V_k
PW_j, RID_j	password, identity of U_j
PID_k, TID_k	pseudo identity, temporary identity of V_k
$s, T_s(x)$	TA's master private key, public key
SC_k, SC_j	smart card of V_k, U_j
$l_{i,j}$	long-term key shared by U_j and TA
N_k, N_j	real parameters of V_k, U_j
TS_1, TS_2, TS_3, TS_4	timestamps

System Setup. TA chooses a big prime p and two integers $s, x \in Z_p^*$, to evaluate the Chebyshev polynomial $T_s(x) \bmod p$. TA selects hash functions $h_i : \{0, 1\}^* \rightarrow Z_p^*$, $i = 1, 2, 3, 4, 5, 6$. Publish $T_s(x), x, p, h_1, \dots, h_6$ and keep s as private key.

The registration, login, authentication and key agreement phases can be depicted as follows (see Table 2).

4 Insecurity of the Cui et al.'s scheme

4.1 Inadequate stored parameters

In the registration phase, V_k stores $\{A_k, N_k^*, n_0\}$ into the memory of SC_k . We find the stored parameters are inadequate for the later computations.

The identity RID_j of each unmanned aerial vehicle U_j should be stored inside of the memory. Otherwise, V_k cannot finish the computation

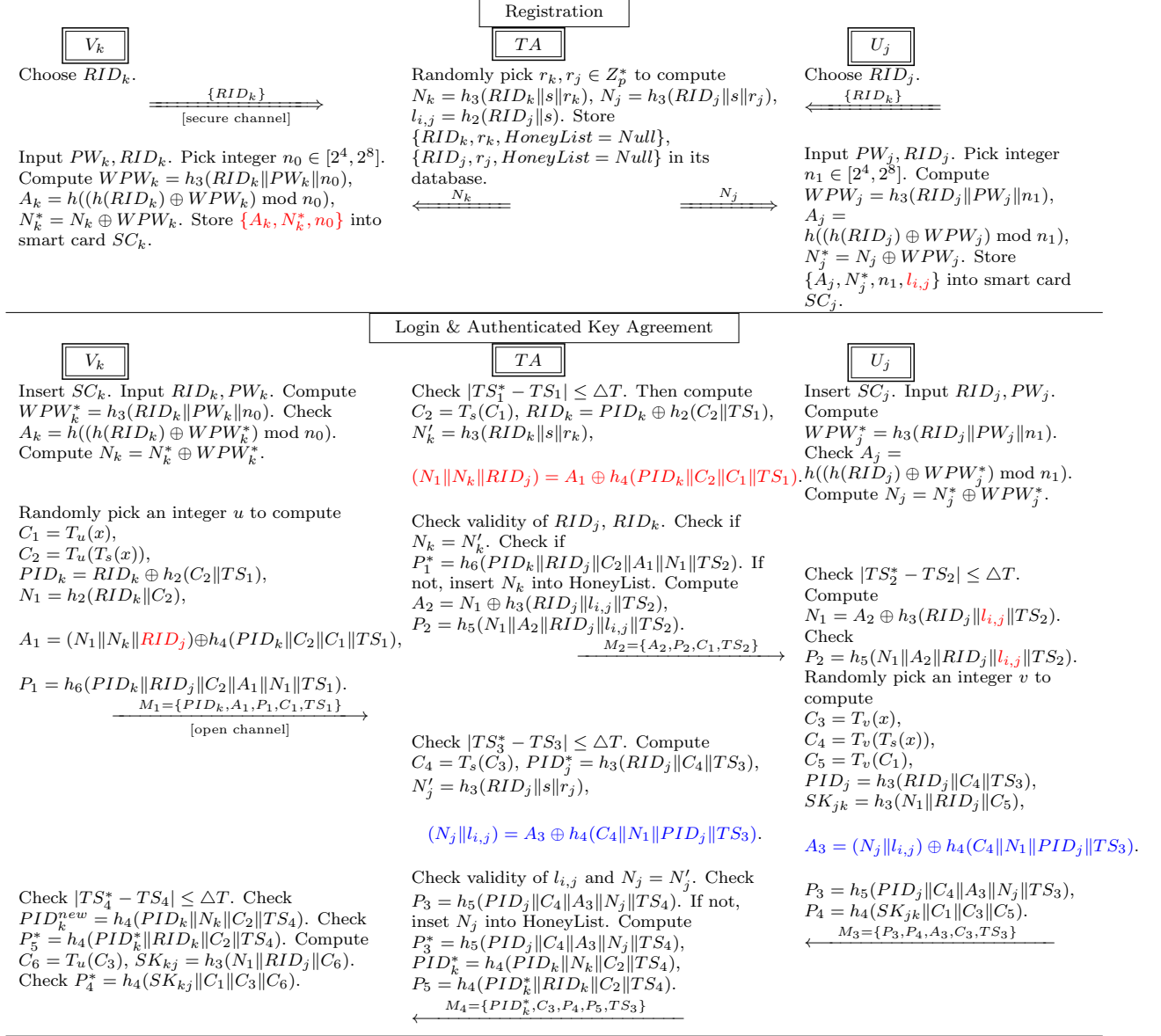
$$A_1 = (N_1 \| N_k \| \textcolor{red}{RID}_j) \oplus h_4(PID_k \| C_2 \| C_1 \| TS_1) \quad (1)$$

So, the actually stored parameters are

$$\{A_k, N_k^*, n_0, \textcolor{blue}{RID}_{j1}, \dots, \textcolor{blue}{RID}_{j\ell}\}$$

where $RID_{j1}, \dots, RID_{j\ell}$ are the identities of unmanned aerial vehicles $U_{j1}, \dots, U_{j\ell}$, which are accessible to V_k .

Table 2: The Cui et al.'s key agreement



4.2 Insecure against ephemeral secret leakage attack

It argues that: “Suppose the $\mathcal{A}$ obtains the common session key $SK_{kj} = h_3(N_1 \| RID_j \| C_6)$ and the ephemeral secrets u and v . However, N_1 is not known by (to) $\mathcal{A}$. Therefore, the proposed scheme can resistance of (resist) ephemeral secret leakage attacks.” (see page 1464, [2]). We find it is not sound.

The scheme uses the operation XOR for encryption and decryption. Concretely,

$$\begin{aligned} \text{[Encryption]} \quad A_1 &= (N_1 \| N_k \| RID_j) \oplus h_4(PID_k \| C_2 \| C_1 \| TS_1), \\ \text{[Decryption]} \quad (N_1 \| N_k \| RID_j) &= A_1 \oplus h_4(PID_k \| C_2 \| C_1 \| TS_1), \end{aligned}$$

where $N_k = h_3(RID_k \| s \| r_k)$. Since $h_i : \{0, 1\}^* \rightarrow Z_p^*$, $i = 1, 2, 3, 4, 5, 6$, we know hash functions h_3 and h_4 have the same output length. Hence, we have

$$A_1 = N_1 \| ((N_k \| RID_j) \oplus h_4(PID_k \| C_2 \| C_1 \| TS_1)) \quad (1')$$

namely, N_1 is entirely copied into the string A_1 . The adversary $\mathcal{A}$ who has intercepted A_1 can retrieve N_1 .

The identity RID_j is shared by vehicles. Once a vehicle is compromised, RID_j is also accessible to the adversary $\mathcal{A}$. Using the ephemeral secret u and intercepted parameter C_3 via the open channel, $\mathcal{A}$ can compute

$$C_6 = T_u(C_3), \quad SK_{kj} = h_3(N_1 \| RID_j \| C_6).$$

Hence, the session key SK_{kj} is eventually exposed.

4.3 Insecure against smart card loss attack

It argues that: “Supposing that $\mathcal{A}$ obtains smart card and extracts $\{A_k, N_k^*, n_0\}$, due to the HoneyList, it is computationally impossible for the $\mathcal{A}$ to proceed without knowing RID_k and PW_k . Therefore, the proposed scheme can resistance (resist) smart card loss attacks.” (see page 1464, [2]). We find the argument is misleading. It only considers the smart card SC_k is compromised, not takes into account that the smart card SC_j is compromised.

First of all, RID_j is accessible to the adversary $\mathcal{A}$ because it is shared by vehicles. Secondly, the encryption and decryption mechanism

$$\begin{aligned} A_3 &= (N_j \| l_{i,j}) \oplus h_4(C_4 \| N_1 \| PID_j \| TS_3), \\ (N_j \| l_{i,j}) &= A_3 \oplus h_4(C_4 \| N_1 \| PID_j \| TS_3) \end{aligned}$$

cannot protect N_j . Since $l_{i,j} = h_2(RID_j \| s)$, and h_2, h_4 have the same output length, the adversary $\mathcal{A}$ who has intercepted A_3 via the open channel, can retrieve N_j because

$$A_3 = N_j \| (l_{i,j} \oplus h_4(C_4 \| N_1 \| PID_j \| TS_3)) \quad (2)$$

Thirdly, the long-term key $l_{i,j}$ is stored inside of SC_j . Once the smart card is cracked, the key will be retrieved. Having the parameters $RID_j, l_{i,j}, N_j$, the adversary $\mathcal{A}$ can impersonate U_j to cheat other vehicles. For example, $\mathcal{A}$ can do as follows (see Table 3).

Table 3: Smart card loss attack against the Cui et al.'s scheme

Upon receiving $M_2 = \{A_2, P_2, C_1, TS_2\}$ from TA, the adversary $\mathcal{A}$ uses the system's public parameters $x, T_s(x)$, and picks an integer $\hat{v}$ to compute $\hat{C}_3 = T_{\hat{v}}(x)$, $\hat{C}_4 = T_{\hat{v}}(T_s(x))$, $\hat{C}_5 = T_{\hat{v}}(C_1)$, $\widehat{PID}_j = h_3(RID_j \parallel \hat{C}_4 \parallel TS_3)$, $SK_{jk} = h_3(N_1 \parallel RID_j \parallel \hat{C}_5)$, $\hat{A}_3 = (N_j \parallel l_{i,j}) \oplus h_4(\hat{C}_4 \parallel N_1 \parallel \widehat{PID}_j \parallel TS_3)$, where TS_3 is the current timestamp. Compute $\hat{P}_3 = h_5(\widehat{PID}_j \parallel \hat{C}_4 \parallel \hat{A}_3 \parallel N_j \parallel TS_3)$, $\hat{P}_4 = h_4(SK_{jk} \parallel \hat{C}_1 \parallel \hat{C}_3 \parallel \hat{C}_5)$. Send $\hat{M}_3 = \{\hat{P}_3, \hat{P}_4, \hat{A}_3, \hat{C}_3, TS_3\}$ to TA. The message $\hat{M}_3$ will pass the TA's verification process.
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4.4 Further discussions

The password PW_j and identity RID_j are always concatenated to compute the stored parameters A_j, N_j^* . The simple setting seems insufficient for resisting smart card loss attack. In our opinion, one should develop other registration mechanism [1] and pseudo-identity updating techniques [3] for the considered scenario.

5 Conclusion

We show that the Cui et al.'s key agreement scheme is insecure against ephemeral secret leakage attack and smart card loss attack. We hope the findings in this note could be helpful for the future work on designing such schemes.

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